An Lam

(832) 421-2248 • anlqt89@gmail.com • Grapevine, Texas • GitHub • LinkedIn • Portfolio

Education

Master of Science in Computer Science, *University of Texas at Arlington* **GPA: 4.00 - Dec 2026**
Bachelor of Science in Computer Science, *Texas Tech University* **GPA: 3.96 - Dec 2022**

Skills

Machine Learning & Artificial Intelligent Classification & Regression, Linear Models & Regularization, Probabilistic Modeling (MLE, MAP), SVM, k-NN, Clustering, Model Evaluation, Search Strategies, Support Vector Machine, Neural Networks, Search & Game Playing Algorithms, Knowledge and Reasoning Logic
Mathematical Foundations Linear Algebra, Probability & Statistics, Optimization (Gradient-Based Methods)
Data & Systems ETL Pipelines, SQL, PostgreSQL, Non-SQL, Modeling Text Data, Exploratory Data Analysis, Dimensional Reduction, Feature Selection
Programming & Software Development Python (Pandas, NumPy, scikit-learn, TensorFlow), Tableau, React, NodeJs, Docker, SQL, C/C++, JavaScript

Work Experience

Database Administrator | PostgreSQL, Python, React **Jan 2023 — Mar 2025**
Phuong Tran Inc *Mentor, Ohio*
- **Architected a PostgreSQL data warehouse** with automated **ETL pipelines** for BI reporting.
- **Developed web scrapers** (Selenium) for competitor tracking, enabling **dynamic pricing** strategies.
- **Built a React/Node.js scheduling system**, reducing idle time by **1hr/day** and increasing revenue by **15%**.
- **Optimized SQL performance** using EXPLAIN ANALYZE to support data-driven decision-making.

Student Researcher | C/C++, Bash, RISC-V **Jan 2022 — Dec 2022**
Texas Tech University *Lubbock, Texas*
- **Engineered automated benchmarking pipelines** in Bash for large-scale RISC-V data collection and analysis.
- **Developed visualization dashboards** to identify bottlenecks and accelerate compiler research diagnostics.

Projects

AI - 8 Puzzles Problem with Search Algorithms **Feb 2026**
*Python, State Space Search, BFS, DFS, DLS, IDS, UCS, Greedy, A**
- **8-Puzzle AI Solver:** Developed a Python-based solver utilizing **BFS, DFS, IDS, DLS, UCS, Greedy,** and **A*** search algorithms to evaluate state-space exploration.
- **Heuristic Optimization:** Implemented **Manhattan L1 distance** for A* search, significantly reducing node expansion and optimizing pathfinding efficiency.
- **Complexity Analysis:** Benchmarked execution time and memory usage across all 7 techniques to analyze **Big-O trade-offs** between completeness and optimality.
- **Robust Logic:** Integrated **inversion-count algorithms** to detect unsolvable configurations and utilized efficient data structures for state-space management.

Vector Space Search Engine **Feb 2026**
Python, Information Retrieval, Sparse Representations, TF-IDF
- Designed a document retrieval system using **sparse TF-IDF vectorization** and **cosine similarity** successfully retrieve the matching documents from search query.
- Performed large-scale text preprocessing and **signal extraction** (tokenization, stemming, normalization).
- Evaluated ranking performance using similarity metrics and **retrieval experiments**.

Movies Big Data Lab **Jan 2025**
PostgreSQL, Python, React, Docker, CI/CD
- Engineered a production-grade analytics platform handling **29M+ records** with optimized relational schema and query pipelines.
- Achieved **5x query speedup** by implementing advanced **indexing strategies** and **materialized views** for large-scale datasets.
- Streamlined deployment using **Docker containerization** and **automated CI/CD pipelines** for reliable cloud-based production releases.

Biomedical Signal Classification **Mar 2024**
Python, Time-Series Modeling, HMM, Random Forest
- Conducted **EDA** on physiological time-series signals (MIT-BIH ECG) to extract discriminative features.
- Built and evaluated **classification models** (HMM, Random Forest) achieving **96% accuracy**.
- Designed **feature pipelines** using sliding-window and FFT-based signal representation in a **healthcare** context.